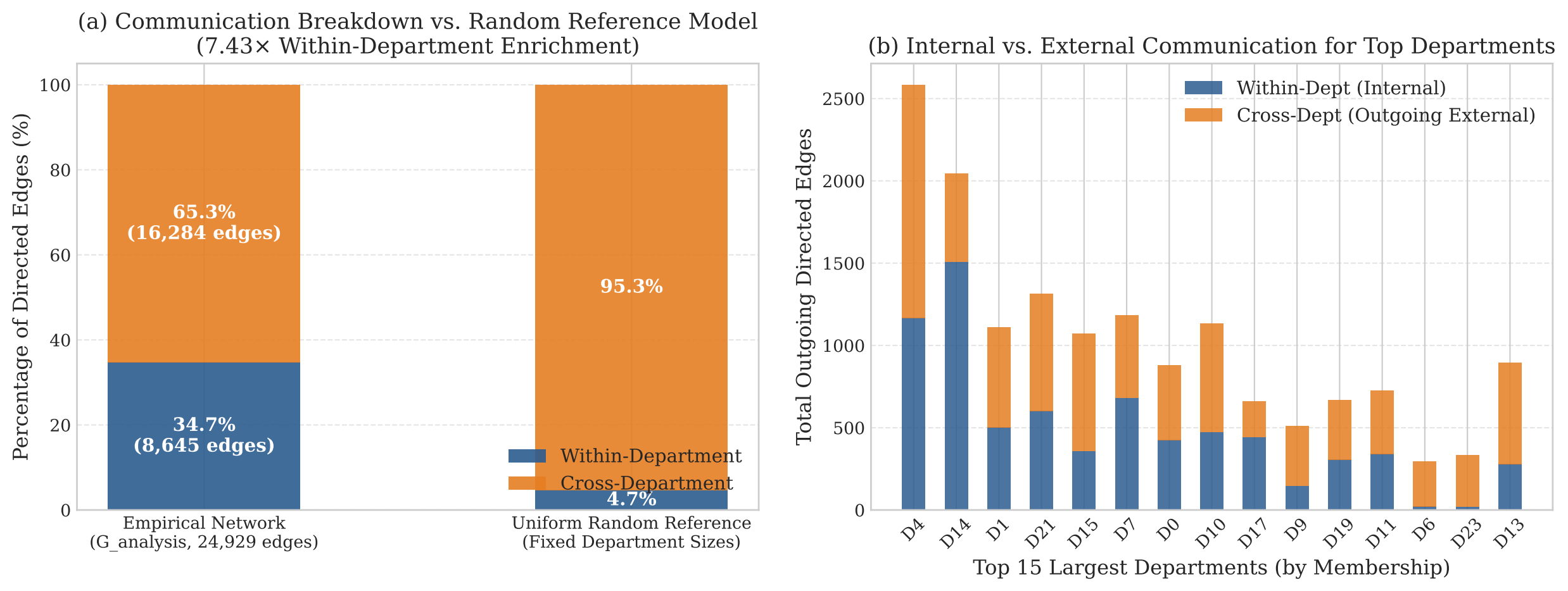
(a) Communication Breakdown vs. Random Reference Model
(7.43× Within-Department Enrichment)
Percentage of Directed Edges (%)
100
80
60
40
20
0
65.3%
(16,284 edges)
34.7%
(8,645 edges)
95.3%
4.7%
Within-Department
Cross-Department
Empirical Network
(G_analysis, 24,929 edges)
Uniform Random Reference
(Fixed Department Sizes)
(b) Internal vs. External Communication for Top Departments
Total Outgoing Directed Edges
2500
2000
1500
1000
500
0
Within-Dept (Internal)
Cross-Dept (Outgoing External)
D4
D14
D1
D21
D15
D7
D0
D10
D17
D9
D19
D11
D6
D23
D13
Top 15 Largest Departments (by Membership)